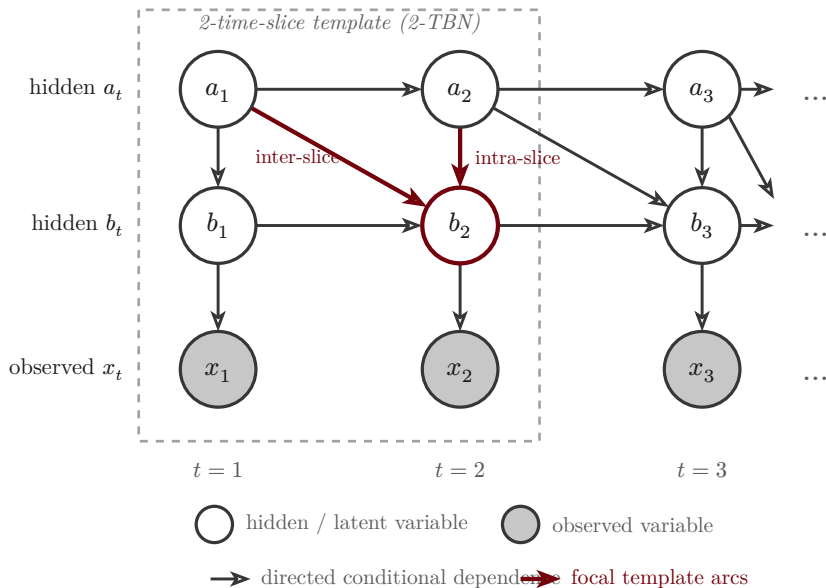


Dynamic Bayesian network — 2-TBN unrolled in time

directed, generative: a repeating 2-slice template copied across t — multiple hidden variables per slice generalise the HMM



$$P(a_{1:T}, b_{1:T}, x_{1:T}) = P(a_1)P(b_1 | a_1)P(x_1 | b_1) \prod_{t \geq 2} P(a_t | a_{t-1}) P(b_t | b_{t-1}, a_{t-1}, a_t) P(x_t | b_t)$$